REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

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Practice Questions and Worked Answers

Grade 9 Advanced | Module 1 Lesson 1 | Methods of Science

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Compare evidence, repeat trials and change only one independent variable.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 3–8.

P01. Which statement best describes a scientific hypothesis?

A. An unchangeable belief B. A testable proposed explanation
C. A measurement unit D. A final proof

Answer

B. A testable proposed explanation.

P02. Why do scientists repeat experiments?

A. To guarantee a preferred answer B. To avoid collecting data
C. To check whether results are reproducible D. To remove the need for controls

Answer

C. To check whether results are reproducible.

P03. What is the main purpose of a scientific model?

A. To reproduce every detail exactly B. To replace all observations
C. To prove that testing is unnecessary D. To represent and explain selected features of a system

Answer

D. To represent and explain selected features of a system.

P04. A scientific theory is best described as:

A. An explanation supported by a broad body of evidence B. An unsupported guess
C. A single experimental reading D. A mathematical unit

Answer

A. An explanation supported by a broad body of evidence.

P05. What should a scientist do when reliable evidence contradicts a model?

A. Hide the evidence B. Reconsider or revise the model
C. Change the measurements without justification D. Stop all investigation

Answer

B. Reconsider or revise the model.

P06. A class tests 4 identical ramps and repeats each test 5 times. How many measurements are collected?

A. 40 measurements B. 10 measurements
C. 20 measurements D. -20 measurements

Answer

C. 20 measurements. Total = number of ramps × repeats = 4 × 5 = 20 measurements.

P07. Three repeated measurements are 12, 14 and 16 s. Calculate their mean.

A. 28 s B. 7 s
C. -14 s D. 14 s

Answer

D. 14 s. Mean = (12 + 14 + 16)/3 = 14 s. Averaging can reduce random variation.

P08. A control test takes 26 s and a test after changing one factor takes 10 s. By how many seconds did the measured time decrease?

A. 16 s B. 32 s
C. 8 s D. -16 s

Answer

A. 16 s. Decrease = control time − changed time = 26 − 10 = 16 s.

P09. An investigation compares 4 surface materials using 5 repeat trials on each. How many trials are needed in total?

A. 40 trials B. 20 trials
C. 10 trials D. -20 trials

Answer

B. 20 trials. Trials = materials × repeats = 4 × 5 = 20.

P10. Repeated readings are 22, 24 and 26 cm. Calculate the range of the readings.

A. 8 cm B. 2 cm
C. 4 cm D. -4 cm

Answer

C. 4 cm. Range = largest − smallest = 26 − 22 = 4 cm.

P11. State the main subject matter of physics.

Answer

Physics investigates matter, energy, motion and their interactions in the physical world.

P12. Distinguish an observation from an inference.

Answer

An observation is information obtained through senses or instruments; an inference is an interpretation based on observations.

P13. Distinguish a scientific law from a scientific theory.

Answer

A law describes a consistent pattern in nature; a theory explains phenomena using a well-supported framework. A theory does not simply become a law.

P14. Why is peer review useful?

Answer

Other scientists examine methods, reasoning and evidence, helping identify weaknesses and improve reliability.

P15. What makes a hypothesis scientifically useful?

Answer

It must lead to predictions that can be tested with observable evidence.

P16. A class tests 4 identical ramps and repeats each test 6 times. How many measurements are collected?

Answer

Total = number of ramps × repeats = 4 × 6 = 24 measurements.

P17. A class tests 4 identical ramps and repeats each test 7 times. How many measurements are collected?

Answer

Total = number of ramps × repeats = 4 × 7 = 28 measurements.

P18. Three repeated measurements are 13, 15 and 17 s. Calculate their mean.

Answer

Mean = (13 + 15 + 17)/3 = 15 s. Averaging can reduce random variation.

P19. Three repeated measurements are 14, 16 and 18 s. Calculate their mean.

Answer

Mean = (14 + 16 + 18)/3 = 16 s. Averaging can reduce random variation.

P20. A control test takes 29 s and a test after changing one factor takes 11 s. By how many seconds did the measured time decrease?

Answer

Decrease = control time − changed time = 29 − 11 = 18 s.

P21. A control test takes 32 s and a test after changing one factor takes 12 s. By how many seconds did the measured time decrease?

Answer

Decrease = control time − changed time = 32 − 12 = 20 s.

P22. An investigation compares 5 surface materials using 5 repeat trials on each. How many trials are needed in total?

Answer

Trials = materials × repeats = 5 × 5 = 25.

P23. An investigation compares 6 surface materials using 5 repeat trials on each. How many trials are needed in total?

Answer

Trials = materials × repeats = 6 × 5 = 30.

P24. Repeated readings are 23, 25 and 27 cm. Calculate the range of the readings.

Answer

Range = largest − smallest = 27 − 23 = 4 cm.

P25. Repeated readings are 24, 26 and 28 cm. Calculate the range of the readings.

Answer

Range = largest − smallest = 28 − 24 = 4 cm.

P26. A student tests whether ramp height changes travel time, but changes both height and surface material. Evaluate the method.

Answer

The effects cannot be separated because two factors changed. Change height only; keep the surface, cart and release method constant.

P27. A student says one result proves a hypothesis forever. Evaluate the claim.

Answer

One result can support a hypothesis but cannot establish it permanently. Repetition, independent tests and consideration of contrary evidence are needed.

P28. Design a fair test of how surface roughness affects the distance a toy car travels.

Answer

Use the same car, release height and starting method. Change only surface roughness, measure travel distance, repeat trials and compare mean distances.

P29. Two groups obtain different results using the same method. Suggest two constructive next steps.

Answer

Check instrument calibration and whether variables were controlled; repeat trials and compare uncertainty. Share methods and data rather than assuming either group is wrong.

P30. A model predicts the correct trend but not the exact measured values. Must it be discarded immediately? Explain.

Answer

Not necessarily. Check uncertainty, assumptions and the model's range of validity. Revise it if discrepancies are systematic and exceed reasonable uncertainty.